UDP SERVER

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/socket.h>
#include <arpa/inet.h>
int main(void)
{
   int server;
   struct sockaddr_in servAddr, client_addr;
   char servMsg[2000], cliMsg[2000];
   socklen_t client_struct_length = sizeof(client_addr);

   server = socket(AF_INET, SOCK_DGRAM, IPPROTO_UDP);
   if (server < 0)
   {
      printf("error while creating socket\n");
      exit(1);
   }
   printf("socket created successfully \n");

   servAddr.sin_family = AF_INET;
   servAddr.sin_port = htons(2002);
   servAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

   if (bind(server, (struct sockaddr*)&servAddr, sizeof(servAddr)) < 0)
   {
      printf("couldn't bind to the port\n");
      exit(1);
   }
   printf("Binding done\n");
   printf("Listening done...\n");
   if (recvfrom(server, cliMsg, sizeof(cliMsg), 0, (struct sockaddr*)&client_addr,
&client_struct_length) < 0)
   {
      printf("Couldn't receive \n");
      exit(1);
   }
   printf("Message from client: %s\n", cliMsg);
   strcpy(servMsg, cliMsg);

   if (sendto(server, servMsg, strlen(servMsg), 0, (struct sockaddr*)&client_addr,
client_struct_length) < 0)
   {
      printf("Can't send\n");
      exit(1);
   }
   close(server);
   return 0;
}
```

UDP CLIENT

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/socket.h>
#include <arpa/inet.h>

int main(void)
{
   int client;
   struct sockaddr_in servAddr;
   char servMsg[2000], cliMsg[2000];
   socklen_t server_struct_length = sizeof(servAddr);

   client = socket(AF_INET, SOCK_DGRAM, IPPROTO_UDP);
   if (client < 0)
   {
      printf("Error while creating socket\n");
      exit(1);
   }
   printf("socket created successfully\n");

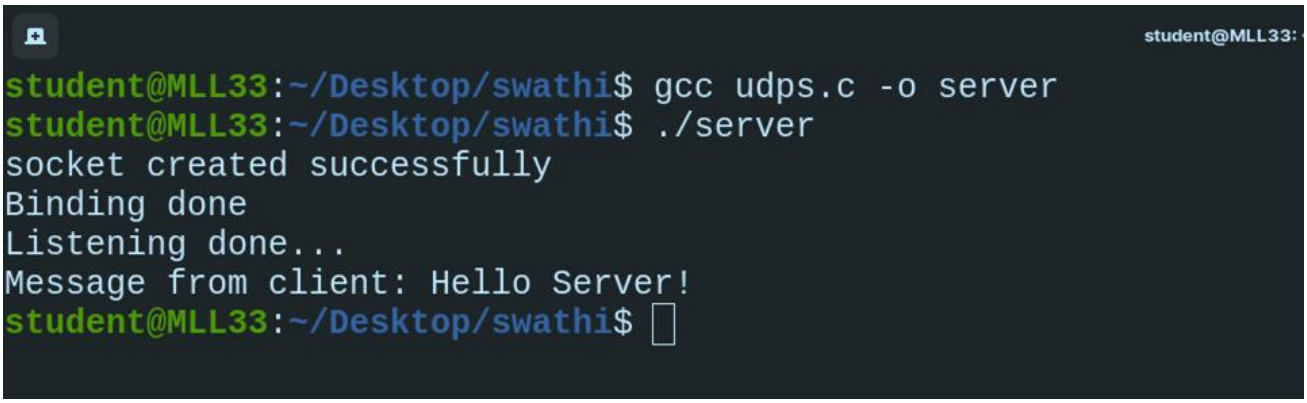
   servAddr.sin_family = AF_INET;
   servAddr.sin_port = htons(2002);
   servAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

   printf("Enter Message to server: ");
   fgets(cliMsg, sizeof(cliMsg), stdin);
   // Remove newline character if present
   cliMsg[strcspn(cliMsg, "\n")] = 0;

   if (sendto(client, cliMsg, strlen(cliMsg), 0, (struct sockaddr*)&servAddr, server_struct_length) <
0)
   {
      printf("unable to send message\n");
      exit(1);
   }

   if (recvfrom(client, servMsg, sizeof(servMsg), 0, (struct sockaddr*)&servAddr,
&server_struct_length) < 0)
   {
      printf("error while receiving server's message\n");
      exit(1);
   }
   printf("Message from server: %s\n", servMsg);
   close(client);
   return 0;
}
```

OUTPUT


student@MLL33:
student@MLL33:~/Desktop/swathi$ gcc udps.c -o server
student@MLL33:~/Desktop/swathi$ ./server
socket created successfully
Binding done
Listening done...
Message from client: Hello Server!
student@MLL33:~/Desktop/swathi$


student@MLL33:
student@MLL33:~/Desktop/swathi$ gcc udpc.c -o client
student@MLL33:~/Desktop/swathi$ ./client
socket created successfully
Enter Message to server: Hello Server!
Message from server: Hello Server!
student@MLL33:~/Desktop/swathi$